

Entropy: Measuring Impurity

In decision tree learning, the algorithm needs a way to measure the "purity" of a set of data.

- A **pure** set is one where all examples belong to the same class (e.g., 100% "Cat" or 100% "Not Cat").
- An **impure** set is a mix of classes.

Entropy, denoted as $H(p_1)$, is a mathematical function used to measure **impurity**.

Understanding Entropy with Examples

Let p_1 be the fraction of positive examples ($y = 1$, e.g., "Cat") in a set.

- **Maximum Impurity (Entropy = 1):**
 - This occurs when the data is a perfect 50/50 split.
 - *Example:* 3 Cats, 3 Dogs $\rightarrow p_1 = 0.5 \rightarrow H(p_1) = 1$.
 - **No Impurity (Entropy = 0):**
 - This occurs when the data is 100% pure (all one class).
 - *Example 1:* 6 Cats, 0 Dogs $\rightarrow p_1 = 1.0 \rightarrow H(p_1) = 0$.
 - *Example 2:* 0 Cats, 6 Dogs $\rightarrow p_1 = 0.0 \rightarrow H(p_1) = 0$.
 - **Some Impurity ($0 < \text{Entropy} < 1$):**
 - This occurs with any other mixed set.
 - *Example 1:* 5 Cats, 1 Dog $\rightarrow p_1 \approx 0.83 \rightarrow H(p_1) \approx 0.65$.
 - *Example 2:* 2 Cats, 4 Dogs $\rightarrow p_1 \approx 0.33 \rightarrow H(p_1) \approx 0.92$.
 - Note that the 2/4 split ($p_1 = 0.33$) is *more impure* (closer to 0.5) than the 5/1 split ($p_1 = 0.83$), so its entropy is higher.
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The Entropy Formula

Let p_1 be the fraction of positive examples and $p_0 = 1 - p_1$ be the fraction of negative examples.

The formula for entropy is:

$$H(p_1) = -p_1 \log_2(p_1) - p_0 \log_2(p_0)$$

or equivalently:

$$H(p_1) = -p_1 \log_2(p_1) - (1 - p_1) \log_2(1 - p_1)$$

Key Conventions:

- **Log Base 2 ($\log_2$):** This is used by convention so that the maximum impurity (at $p_1 = 0.5$) is exactly **1**.
 - **Special Case ($0 \log_2 0$):** Since $\log_2(0)$ is undefined, we use the convention that $0 \cdot \log_2(0) = 0$. This ensures that pure sets ($p_1 = 0$ or $p_0 = 0$) correctly result in an entropy of 0.
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Alternatives to Entropy

- Other measures of impurity exist, such as the **Gini criteria** (or Gini impurity).
- The Gini function looks very similar to the entropy curve and serves the same purpose: to quantify the impurity of a node.
- This course will focus on using entropy.